

Chaitanya Mamatha Ananda

Compiler Optimization Engineer · Ph.D. Candidate · LLVM/BOLT

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Summary

Compiler Ph.D. candidate specializing in post-link and profile-guided optimization for x86 and Arm (AArch64). First-author on **CGO**, **LCTES**, and **CAIS/MLArchSys** publications: heap-layout optimization that reduces execution time by **21.7%** on average, and cross-module basic block deduplication that shrinks binaries by up to **25.8%**. Recently at Google, applied AI-driven search (AlphaEvolve) to warehouse-scale interprocedural code layout.

Technical Skills

Languages & Tools: C++, C, Python, NumPy, Linux, Git

Compilers & Optimization: LLVM, BOLT, Propeller, profile-guided optimization (PGO/FDO), post-link and binary optimization, code and memory layout, basic block reordering, interprocedural optimization

Performance Engineering: x86/Arm64, perf, hardware counters, profiling, warehouse-scale workloads

ML for Compilers: AI-driven heuristic search, AlphaEvolve, Vizier, Evolution Strategies, MLGO

Experience

➤ **Student Researcher** ·  **Google LLC**

Sunnyvale, CA · Sep 2025 – May 2026

AI-driven code layout optimization for warehouse-scale binaries; Manager: Sriraman Tallam

- Extended Google's **Propeller** post-link optimizer with AI-driven interprocedural code layout for warehouse-scale binaries, guided by hardware performance counters.
- Improved execution time by **1.6%** on *LLVM Clang* and **0.23%** on a production *Search* workload.

➤ **Ph.D. Candidate** · **University of California, Riverside**

Riverside, CA · Sep 2022 – Present

Data locality improvement and code footprint reduction; Advisor: Dr. Rajiv Gupta

- **PreFix:** Profiling-guided heap optimization that groups hot objects to improve locality, reducing execution time by **21.7%** on average.
- **DeduBB:** Propeller-based post-link optimizer for x86 and Arm that deduplicates cross-module basic blocks, reducing binary size by up to **25.8%** under PGO.

➤ **Research Assistant** · **Indian Institute of Science**

Bengaluru, India · Aug 2019 – May 2022

Parallel programming and scientific data analysis

- Built a **Regent/Legion** parallel programming model for PDE solvers and designed deep-learning anomaly detectors for scientific data.

➤ **Intern** · **Robert Bosch Engineering and Business Solutions**

Bengaluru, India · Spring 2021

Automated driving data pipeline

- Developed scripts to synchronize video and radar data from automated test driving.

Publications

- **CGO 2025:** *PreFix: Optimizing the Performance of Heap-Intensive Applications*.
C. Mamatha Ananda, R. Gupta, S. Tallam, H. Shen, and X.D. Li.
- **LCTES 2026:** *DeduBB: Binary Code Size Reduction via Post-Link Basic Block Deduplication*.
C. Mamatha Ananda, M. Afarin, R. Gupta, S. Tallam, H. Shen, and X.D. Li.
- **CAIS & MLArchSys 2026:** *AI-PROPELLER: Warehouse-Scale Interprocedural Code Layout Optimization with AlphaEvolve*. (Selected for Oral Presentation).
C. Mamatha Ananda, R. Gupta, M. Trofin, A. Grossman, S. Tallam, X.D. Li, and A. Yazdanbakhsh.

Patents

▪ **Generate Better Code Layout Heuristics** · 

Patent Filed 2026 · Co-inventor

Education

Ph.D. in Computer Science, UC Riverside

Sep 2022 - Present

B.E. in Computer Science, Bangalore Institute of Technology, India

2021

Awards & Teaching

Award: Dean's Distinguished Fellowship, UC Riverside (2022)

Teaching: Compiler Design and Construction.